

EZEKIEL ANDREASSEN

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EXPERIENCE

Smartapp.com | Construction Management Software Company

R&D Hardware Lead - Digital Reconstruction | Smartapp.com | Worcester, MA | Dec 2024 - Present

- Managed optimization of software reconstruction pipeline, reducing runtime from 10 hours to under 2 hours
- Report weekly to executives on program health, surfacing risks and aligning timelines with annual strategic goals

Robotics Engineer II | Oct 2023 - Nov 2024

- Designed new mobile robot platform: reduced robot size by over 75%, increased peak payload capacity by 150%, and enabled field demo of integrated motion, sensing and compute stack.
- Redesigned robotic joints through a new mechanical architecture, custom control and motion kinematics, enabling a **50% reduction in joint size** and a **37% increase in efficiency**; validated the improvements with custom test equipment and demonstrated results in a live internal review.
- Designed, built and tested thermal enclosure & motion system for a combination gimbaled LiDAR/camera

Robotics Engineer I | Oct 2022 - Sep 2023

- Turned a proof of concept design for a human operated mobile robot and made it internally demoable
- Wrote a proposal for a refined system architecture to unify sensing, compute, structure, and actuation — aligning engineering capabilities with business goals and enabling buy-in for next-stage development.

Experiential Robotics | Non Profit Education Startup

Contract R&D Mechatronics Engineer | Manchester, NH | Sep 2023 - Oct 2024

- Designed a low-cost, educational robot with Idaho National Laboratories and Cornell University, with initial testing next semester in five classrooms across Idaho, to modernize high school agriculture classes.
- Executed strategic design decisions to ease sourcing, assembly, and manufacturing constraints by reducing part counts, minimizing BOM changes between SKUs, and lowering dexterity requirements for assembly.

DEKA Research | Robotics & Medical R&D Firm

Contract Mechanical Engineer (Education Department) | Manchester, NH | June 2022 - Oct 2022

- Developed a low-cost, educational robotics kit for middle and high school students to learn programming and gain hands-on robotics experience, leading to the sale of 15,000+ beta units in the first year.
- Designed injection molded parts with weekly design reviews with staff plastics engineers.

PROJECTS

BattleBots | Heavyweight Combat Robotics | May 2019 - Aug 2021

- Top 8 Finisher on 250 lb robot fighting television show with 5 million viewers
- Drove root-cause analysis of wheel failure; redesigned assembly, updated material + fixturing, and produced vendor drawings—new version survived harsher test loads with zero breakage.

All-Terrain Electric Skateboard | Torsion Rod Suspension | Early 2022-ish

- Designed, simulated, built, and tested a torsion rod suspension for an all-terrain electric skateboard; used MATLAB to optimize a non-linear spring linkage for ideal frequency response under dynamic loading conditions.

EDUCATION

WPI | Bachelors of Science in Robotics Engineering | Worcester, MA | May 2022

GPA 3.5/4.0, Rho Beta Epsilon, Unified Robotics Course Staff, Outing Club, Boulderling Gym Manager, E-skate Club

SKILLS

Beginner Sheet Metal | Plastic & Metal Extrusion

Intermediate Injection Molding | Thermal, Static Stress, Modal FEA | GD&T | Ubuntu

Advanced CAD | PDM | Release Control | Python